Problem 7.4: Determine the discharge for system as given in Figure and calculate the pressures at Points A, B, C, D. Head loss per unit length and head loss at the nozzle may be calculated with using equations of $\left(\frac{0.1\,V^2}{2g}\right)$ and $\left(\frac{3V^2}{2g}\right)$, respectively. Neglect the head losses at elbows and entrance.

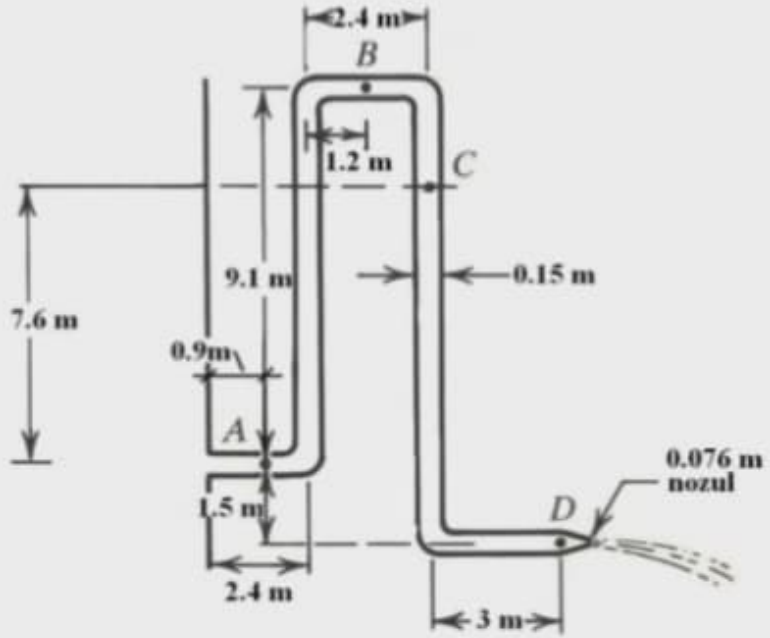


Figure 7.4

(1 – 2) *Enerji Denklemi*

$$\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1=\frac{P_2}{\gamma}+\frac{V_2^2}{2g}+z_2+\sum\Delta h$$

$$9.1=\frac{V_2^2}{2g}+0.1\frac{V_1^2}{2g}(2.4+9.1+2.4+10.6+3)$$

$$+3\frac{V_2^2}{2g}$$

$$9.1=4\frac{V_2^2}{2g}+2.75\frac{V_1^2}{2g}$$

$$\frac{V_1\pi*0.15^2}{4}=\frac{V_2\pi*0.076^2}{4}$$

$$V_1=V_2*0.2567$$

$$9.1 = 4\frac{V_2^2}{2g} + \frac{2.75(V_2 * 0.2567)^2}{2g}$$

$$9.1 = \frac{4.181 * V_2^2}{2g}$$

$$V_2 = 6.53\ m/s\ ;\ V_1 = 1.68\ m/s\ \Rightarrow Q = V * A = 0.0297\ m^3/s$$

(1 – A)

$$\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1=\frac{P_A}{\gamma}+\frac{V_A^2}{2g}+z_A+\sum\Delta h_1$$

$$7.6=\frac{P_A}{\gamma}+\frac{1.68^2}{2g}+0.9*0.1*\frac{1.68^2}{2g}\Rightarrow P_A=74.43\ kN/m^2$$

(1 – B)

$$\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1=\frac{P_B}{\gamma}+\frac{V_B^2}{2g}+z_B+\sum\Delta h_2$$

$$0=\frac{P_B}{\gamma}+\frac{1.68^2}{2g}+1.5+(2.4+9.1+1.2)*0.1*\frac{1.68^2}{2g}\Rightarrow P_B=-18.27\ kN/m^2$$

(1-C)

$$\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1=\frac{P_C}{\gamma}+\frac{V_C^{\,2}}{2g}+z_C+\sum\Delta h_3$$

$$0=\frac{P_B}{\gamma}+\frac{1.68^2}{2g}+(2.4+9.1+2.4+1.5)*0.1*\frac{1.68^2}{2g}\ \Rightarrow\ P_C=-3.65\ kN/m^2$$

(1-D)

$$\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1=\frac{P_D}{\gamma}+\frac{V_D^{\,2}}{2g}+z_D+\sum\Delta h_4$$

$$9.1=\frac{P_D}{\gamma}+\frac{1.68^2}{2g}+(2.4+9.1+2.4+1.5+9.1+3)*0.1*\frac{1.68^2}{2g}\ \Rightarrow\ P_D=85.6\ kN/m^2$$

Problem 7.5: A fluid with a kinematic viscosity $\nu=15\times10^{-7}$ m²/s flows in the pipe system in Figure 7.5. The roughness of the pipe is k=0.001 m and the minor head loss coefficients of the system are known as $K_{contraction}=0.21$ $K_{entrance}=0.05$. Find the discharge of the system by using Moody diagram.

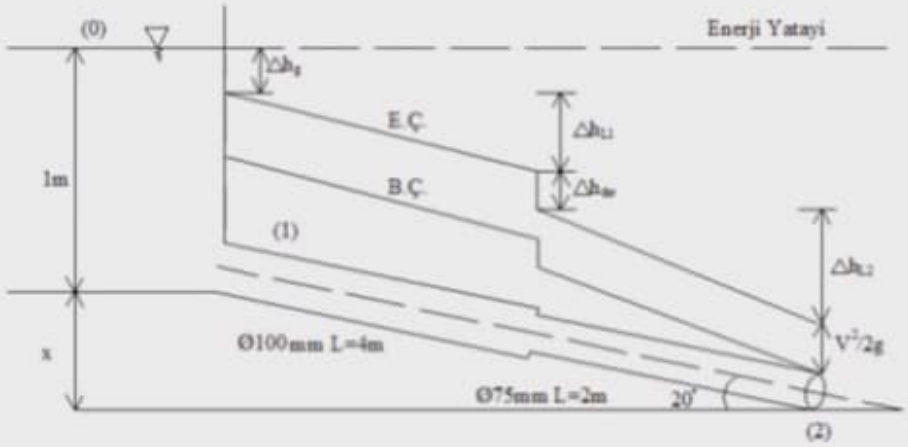


Figure 7.5

Solution:

If the energy equation between the reservoir and outlet of the pipe is written (0-2)

$$\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_2}{\gamma}+\frac{V_2^2}{2g}+z_2+\sum\Delta h$$

$$1+x=\frac{V_2^2}{2g}+\sum\Delta h$$

$x=6\times\sin 20=2.05$ m

$$3.05=\frac{V_2^2}{2g}+K_g\frac{V_1^2}{2g}+\lambda\frac{L_1}{D_1}\frac{V_1^2}{2g}+K_{dar}\frac{V_2^2}{2g}+\lambda\frac{L_2}{D_2}\frac{V_2^2}{2g}$$

Continuity equation

$Q_1=Q_2$ $\quad$ $V_1\times A_1=V_2\times A_2$

$V_2=1.78\ V_1$

By using Moody diagram let's assume $\lambda=0.02$

$$3.05=\frac{(1.78V_1)^2}{2g}+0.05\times\frac{V_1^2}{2g}+0.02\times\frac{4}{0.1}\times\frac{V_1^2}{2g}+0.21\times\frac{(1.78V_1)^2}{2g}+0.02\times\frac{2}{0.075}\times\frac{(1.78V_1)^2}{2g}$$

$$3.05=3.88\times\frac{V_1^2}{2g}+2.49\times\frac{V_1^2}{2g}$$

$$3.05 = 6.37 \times \frac{V_1^2}{2g} \qquad V_1 = 3.06 \text{ m/s}$$

in order to control

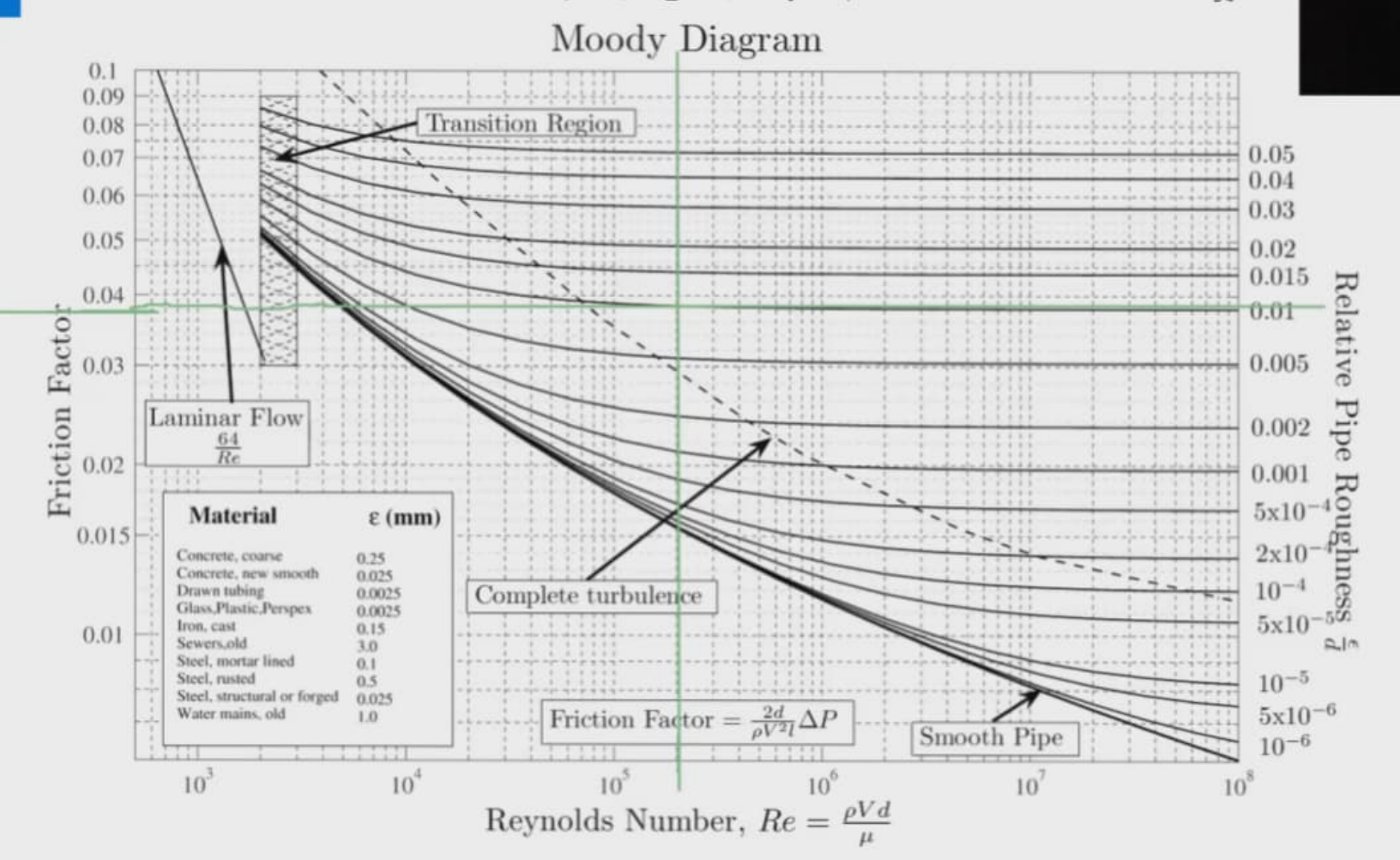
$$Re_1 = \frac{3.06 \times 0.1}{15 \times 10^{-7}} = 204333$$

$$\frac{D_1}{k} = \frac{0.1}{0.001} = 100$$

$\lambda = 0.037$ is obtained but this value is different from the first assumption, so

Assume a new $\lambda = 0.037$

$$3.05 = 3.88 \times \frac{V_1^2}{2g} + 4.61 \times \frac{V_1^2}{2g}$$


Moody Diagram
Transition Region
Laminar Flow
$\frac{64}{Re}$
Complete turbulence
Friction Factor = $\frac{2d}{\rho V^2 l}\Delta P$
Smooth Pipe
Material
ε (mm)
Concrete, coarse 0.25
Concrete, new smooth 0.025
Drawn tubing 0.0025
Glass,Plastic,Perspex 0.0025
Iron, cast 0.15
Sewers,old 3.0
Steel, mortar lined 0.1
Steel, rusted 0.5
Steel, structural or forged 0.025
Water mains, old 1.0
Friction Factor
Reynolds Number, $Re = \frac{\rho V d}{\mu}$
Relative Pipe Roughness $\frac{\varepsilon}{d}$
0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.015
0.01
0.05
0.04
0.03
0.02
0.015
0.01
0.005
0.002
0.001
$5x10^{-4}$
$2x10^{-4}$
10^{-4}
$5x10^{-5}$
10^{-5}
$5x10^{-6}$
10^{-6}
10^3
10^4
10^5
10^6
10^7
10^8

$$3.05 = 8.49 \times \frac{V_1^2}{2g} \qquad V_1 = 2.65 \text{ m/s}$$

$$Re_1 = \frac{2.65 \times 0.1}{15 \times 10^{-7}} = 176667$$

$$\frac{D}{k} = 100$$

λ=0.037 O.K.

V_2=4.72 m/s

$$Re_2=\frac{4.72\times 0.075}{15\times 10^{-7}}=235850$$

$$\frac{D}{k}=\frac{0.075}{0.001}=75$$

λ=0.038 O.K

So

$$Q=V_1\times A_1=2.65\times\frac{\pi\times 0.1^2}{4}=0.0206\ m^3/s$$

Problem 7.6: Water flows from the reservoir A to reservoir B in the Figure 7.6. The elevation difference between the reservoirs is Δh=5.00 m. Find the discharge. Draw the energy and hydraulic grade line. (g=10 m/s^2, ν=10^{-6} m^2/s)

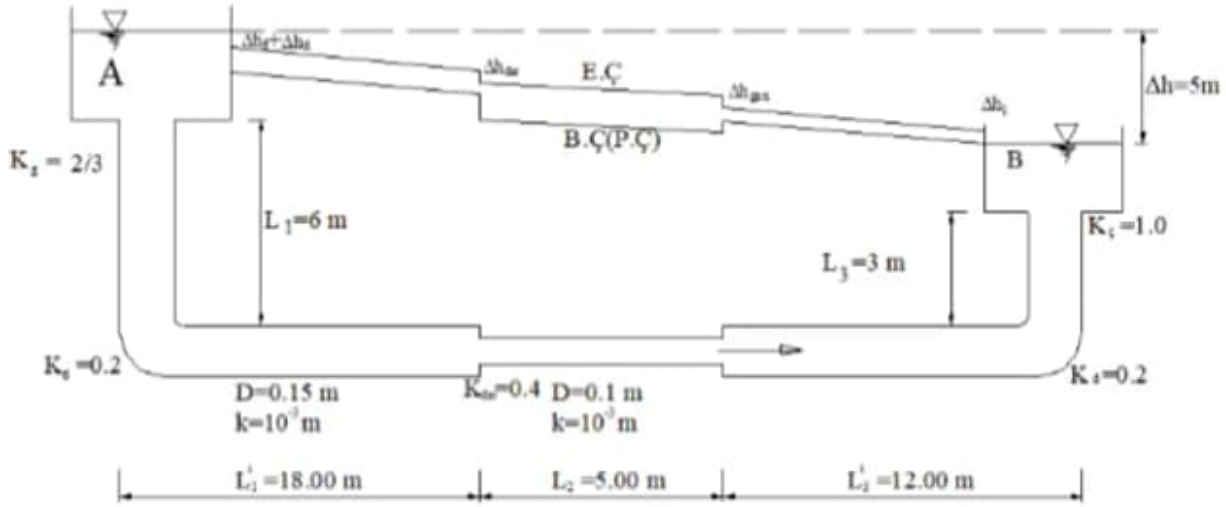


Figure 7.6

Solution:

$$\frac{P_A}{\gamma}+\frac{V_A^2}{2g}+z_A=\frac{P_B}{\gamma}+\frac{V_B^2}{2g}+z_B+\sum\Delta h_{AB}$$

$$z_A-z_B=\sum\Delta h_{AB}=K_g\frac{V_1^2}{2g}+\lambda_1\frac{L_1}{D_1}\frac{V_1^2}{2g}+K_d\frac{V_1^2}{2g}+K_{dar}\frac{V_2^2}{2g}+\lambda_2\frac{L_2}{D_2}\frac{V_2^2}{2g}+\frac{(V_2-V_3)^2}{2g}$$

$$+\lambda_3\frac{L_3}{D_3}\frac{V_3^2}{2g}+K_d\frac{V_3^2}{2g}+K_c\frac{V_3^2}{2g}$$

$$C=18\log\frac{6R}{a+\frac{\delta_*}{7}}\qquad\qquad\lambda=\frac{8g}{C^2}$$

Let's assume hydraulically rough and ignore $\frac{\delta_*}{7}$.

$$C=18\log\frac{6R}{a}$$

$$C = 18\log\frac{6R}{a}$$

$$C_1 = 18\log\frac{6(\frac{0.15}{4})}{\frac{10^{-3}}{2}} = 47.76\ m^{1/2}/s \qquad \lambda_1 = \frac{8\times 9.81}{47.76^2} = 0.035$$

$$C_2=18\log\frac{6\times(\frac{0.1}{4})}{\frac{10^{-3}}{2}}=44.59\ m^{1/2}/s \qquad \lambda_2=0.04$$

$C_3=C_1=44.76\ m^{1/2}/s$ $\lambda_3=\lambda_1=0.035$

From the continuity equation $Q_1=Q_2=Q_3$

$V_1A_1=V_2A_2$ $V_1\times\frac{\pi\times0.15^2}{4}=V_2\times\frac{\pi\times0.1^2}{4}$ $V_2=2.25\ V_1$ $V_3=V_1$

$$5 = 0.25\frac{V_1^2}{2g} + 0.035\times\frac{24}{0.15}\times\frac{V_1^2}{2g} + 0.2\times\frac{V_1^2}{2g} + 0.4\times\frac{5V_1^2}{2g} + 0.04\times\frac{5}{0.1}\times\frac{5V_1^2}{2g} + 1.25^2\times\frac{V_1^2}{2g}$$
$$+ 0.035\times\frac{15}{0.15}\times\frac{V_1^2}{2g} + 0.2\times\frac{V_1^2}{2g} + \frac{V_1^2}{2g}$$

V_1=2.02 m/s

$$Q = 2.02\times\frac{\pi\times 0.15^2}{4} = 0.0357\ m^3/s$$

V_2=4.5 m/s

Control

$$\delta_{*1} = \frac{11.6\gamma}{u_{*1}}$$

$$J_1 = \lambda_1 \frac{1}{D_1} \frac{V_1^2}{2g} = 0.035 \times \frac{1}{0.15} \times \frac{2.02^2}{19.62} = 0.048$$

$$u_{*1} = \sqrt{gRJ} = \sqrt{10 \times \frac{0.15}{4} \times 0.048} = 0.135 \text{ m/s}$$

$$\delta_{v1} = \frac{11.6 \times 10^{-6}}{0.135} = 8.6 \times 10^{-5}$$

$$\frac{\delta_{v1}}{7} = 1.23 \times 10^{-5} m << k = \frac{10^{-3}}{\square} = 10^{-3} m$$ Hydraulically rough

$$J_2 = 0.04 \times \frac{1}{0.1} \times \frac{4.5^2}{19.62} = 0.412$$

$$u_{*2} = \sqrt{10 \times \frac{0.1}{4} \times 0.412} = 0.45 \text{ m/s}$$

$$\delta_{v2} = \frac{11.6 \times 10^{-6}}{0.45} = 2.55 \times 10^{-5}$$

$$\frac{\delta_{v2}}{7} = 3.65 \times 10^{-6} \text{ m} << k = 5 \times 10^{-4} \text{ m}$$ Hydraulically rough

assumption is correct